

Math 526, S06
Exam 2

Name:
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
Total	100	

Good Luck !

Problem 1. (20 points) Let $\mathbf{x} = \begin{bmatrix} 1 \\ -6 \\ 4 \\ 2 \end{bmatrix}$, $\mathbf{A} = \begin{bmatrix} 2 & 4 & 0 \\ -1 & -2 & 5 \\ 2 & 4 & -5 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ and

$$\mathbf{C} = \begin{bmatrix} -2 & 3 \\ 2 & -1 \end{bmatrix}.$$

a) Compute $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$, and $\|\mathbf{x}\|_\infty$.

Solution:

$$\|\mathbf{x}\|_1 = 1 + 6 + 4 + 2 = 11$$

$$\|\mathbf{x}\|_2 = \sqrt{1 + (-6)^2 + 4^2 + 2^2} = \sqrt{57}$$

$$\|\mathbf{x}\|_\infty = \max\{2, 6, 4, 2\} = 6$$

b) Compute $\|\mathbf{A}\|_1$ and $\|\mathbf{A}\|_\infty$.

Solution:

$$\|\mathbf{A}\|_1 = \max\{2 + 1 + 2, 4 + 2 + 4, 0 + 5 + 5\} = 10$$

$$\|\mathbf{A}\|_\infty = \max\{2 + 4 + 0, 1 + 2 + 5, 2 + 4 + 5\} = 11$$

b) Compute $\|\mathbf{B}\|_2$.

Solution:

$$\|\mathbf{A}\|_1 = \max\{2, 5, 4\} = 5$$

c) Compute $\kappa_\infty(\mathbf{C})$.

Solution:

$$\mathbf{C}^{-1} = \begin{bmatrix} 1/4 & 3/4 \\ 1/2 & 1/2 \end{bmatrix}$$

So

$$\kappa_\infty(\mathbf{C}) = \|\mathbf{C}\|_\infty \cdot \|\mathbf{C}^{-1}\|_\infty = 5 \cdot 1 = 5$$

Problem 2. (20 points) Consider the system $\mathbf{Ax} = \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 4 & -2 & 0 \\ 1 & 4 & 1 \\ -2 & 1 & -5 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -6 \\ 6 \\ -12 \end{bmatrix}$$

a) Write down the corresponding iterative formula for the **Gauss-Seidel method**, then compute the approximate solutions $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ of the system by the Gauss-Seidel method with the initial guess $\mathbf{x}_0 = \mathbf{0}$.

Solution: The iterative formula for the Gauss-Seidel method is

$$\begin{aligned} x_1^{new} &= \frac{1}{4}(-6 + 2x_2^{old}) \\ x_2^{new} &= \frac{1}{4}(6 - x_1^{new} - x_3^{old}) \\ x_3^{new} &= -\frac{1}{5}(-12 + 2x_1^{new} - x_2^{new}) \end{aligned}$$

With $\mathbf{x}_0 = [0 \ 0 \ 0]^t$, we get

$$\begin{aligned} \mathbf{x}_1 &= \begin{bmatrix} -1.5 \\ 1.875 \\ 3.375 \end{bmatrix} \\ \mathbf{x}_2 &\approx \begin{bmatrix} -0.5625 \\ 0.7968 \\ 2.7800 \end{bmatrix} \\ \mathbf{x}_3 &\approx \begin{bmatrix} -1.1015 \\ 1.0803 \\ 3.0567 \end{bmatrix} \end{aligned}$$

b) Compute $\|\mathbf{x}_3 - \mathbf{x}\|_\infty$ where the exact solution is given by $\mathbf{x} = [-1 \ 1 \ 3]^t$?

Solution: For the Gauss-Seidel method, we have

$$\|\mathbf{x} - \mathbf{x}_3\|_\infty \approx 0.1015$$

Problem 3. (20 points) Consider the following equation

$$\begin{bmatrix} 7 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 7 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 7 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 7 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 7 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 7 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 7 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

The **Jacobi method** is applied to solve this system.

a) Compute the convergence rate $\|\mathbf{S}^{-1}\mathbf{T}\|_\infty$.

Solution:

Fy the Jacobi method, we have

$$\mathbf{S}^{-1}\mathbf{T} = \begin{bmatrix} 0 & -1/7 & 0 & 0 & -1/7 & 0 & 0 & 0 \\ 0 & 0 & -1/7 & 0 & 0 & -1/7 & 0 & 0 \\ 0 & 0 & 0 & -1/7 & 0 & 0 & -1/7 & 0 \\ 0 & 0 & 0 & 0 & -1/7 & 0 & 0 & -1/7 \\ -1/7 & 0 & 0 & -1/7 & 0 & 0 & 0 & 0 \\ 0 & -1/7 & 0 & 0 & -1/7 & 0 & 0 & 0 \\ 0 & 0 & -1/7 & 0 & 0 & -1/7 & 0 & 0 \\ 0 & 0 & 0 & -1/7 & 0 & 0 & -1/7 & 0 \end{bmatrix}$$

So

$$\|\mathbf{S}^{-1}\mathbf{T}\|_\infty = \frac{2}{7}$$

b) How many steps are needed to reduce the initial solution error (in the $\|\cdot\|_\infty$ norm) by a factor of 10^{-6} ?

Solution:

We need to solve k for

$$\frac{\|\mathbf{e}_k\|_\infty}{\|\mathbf{e}_0\|_\infty} \leq 10^{-6}$$

where $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}_0$. But we know

$$\frac{\|\mathbf{e}_k\|_\infty}{\|\mathbf{e}_0\|_\infty} \leq \|\mathbf{S}^{-1}\mathbf{T}\|_\infty^k$$

So we only need to solve

$$\frac{\|\mathbf{e}_k\|_\infty}{\|\mathbf{e}_0\|_\infty} \leq (2/7)^k \leq 10^{-6}$$

which gives

$$k \geq \frac{\ln(10^{-6})}{\ln(2/7)} \geq 12$$

Problem 4. (20 points)

a) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & -3 & 1 \\ 1 & 0 & 4 & 2 \\ 0 & 1 & 0 & 0 \\ 2 & 3 & 1 & 3 \end{bmatrix}$. Compute $\det(\mathbf{A})$. Is $\mathbf{A}$ singular?

Solution:

$$\begin{aligned} \det(\mathbf{A}) &= \begin{vmatrix} 1 & 2 & -3 & 1 \\ 1 & 0 & 4 & 2 \\ 0 & 1 & 0 & 0 \\ 2 & 3 & 1 & 3 \end{vmatrix} = (-1) \cdot \det \begin{bmatrix} 1 & -3 & 1 \\ 1 & 4 & 2 \\ 2 & 1 & 3 \end{bmatrix} \\ &= -\left(1 \cdot \det \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} - (-3) \cdot \det \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} + 1 \cdot \det \begin{bmatrix} 1 & 4 \\ 2 & 1 \end{bmatrix}\right) \\ &= -(10 - 3 - 7) \\ &= 0 \end{aligned}$$

Thus $\mathbf{A}$ is singular.

b) Is it correct that $\det \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} = \det \begin{bmatrix} d & b & a & c \\ -h & -f & -e & -g \\ -l & -j & -i & -k \\ p & n & m & o \end{bmatrix}$? Explain.

Solution: Yes.

$$\begin{aligned} \det \begin{bmatrix} d & b & a & c \\ -h & -f & -e & -g \\ -l & -j & -i & -k \\ p & n & m & o \end{bmatrix} &= -\det \begin{bmatrix} d & b & a & c \\ h & f & e & g \\ -l & -j & -i & -k \\ p & n & m & o \end{bmatrix} = \det \begin{bmatrix} d & b & a & c \\ h & f & e & g \\ l & j & i & k \\ p & n & m & o \end{bmatrix} \\ &= -\det \begin{bmatrix} a & b & d & c \\ e & f & h & g \\ i & j & l & k \\ m & n & p & o \end{bmatrix} = \det \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} \end{aligned}$$

Problem 5. (20 points) True or False (no explanation needed).

a) Let $\mathbf{x}$ be a column 4-vector, then $\|\mathbf{x}\|_1 \leq 2\|\mathbf{x}\|_2$.

True

b) Let $\mathbf{A}$ be a 4×4 matrix, then $\kappa_1(\mathbf{A}) \leq 4\kappa_\infty(\mathbf{A})$.

False

c) There is a 5×5 matrix $\mathbf{A}$ such that $\kappa_\infty(\mathbf{A}) = 0.8$.

False

d) Let $\mathbf{A}$ be a nonsingular matrix, then we have $\det(\mathbf{A}) = \frac{1}{\det(\mathbf{A}^{-1})}$.

True

e) The Gauss-Seidel method doesn't converge for the system $\begin{bmatrix} 1 & -0.4 \\ 2 & 1 \end{bmatrix} \mathbf{x} = \mathbf{b}$.

False